

Lecture-20

Chi-Square Based Tests

- A chi sq. test is used when our data is counts/frequencies rather than measurements. Compare what we actually observed with what we expected to observe if H_0 was true.

$$\frac{(O - E)^2}{E}, \quad O \rightarrow \text{obs. freq.}, \quad E \rightarrow \text{expected freq.} \quad \left| \chi^2 = \sum \frac{(O - E)^2}{E} \right|$$

small $\chi^2 \Rightarrow$ good agreement with $H_0 \Rightarrow$ good fit

large $\chi^2 \Rightarrow$ evidence against $H_0 \Rightarrow$ bad fit

- χ^2 goodness of fit test $\rightarrow$ testing whether a population follows specified distribution.

1, 2, ..., k

$p_1, p_2, \dots, p_k$

H_0 : populaⁿ follows distribuⁿ

H_1 : not " "

$$\chi^2 = \sum_{i=1}^k \frac{(O_i - E_i)^2}{E_i} \sim \chi^2_{k-1} \quad \nu = k-1$$

eg. some data

$$H_0: p_1 = p_2 = \dots = p_6 = \frac{1}{6}$$

H_1 : at least one p_i is not $\frac{1}{6}$

$$E_i = \frac{120 \times 1}{6} = 20$$

$$\chi^2 = \frac{0}{20} + \frac{4}{20} + \frac{9}{20} + \frac{4}{20} + \frac{1}{20} + \frac{16}{20} = \frac{34}{20} = 1.7$$

$$\chi^2 \sim \chi^2_5 \quad \alpha = 0.05 \quad \chi^2_{0.05, 5} = 11.07$$

since $1.7 < 11.07 \Rightarrow$ fail to reject H_0 .

- reject H_0 if $\boxed{\chi^2_{\text{calc}} > \chi^2_{\alpha, \nu}}$ always right-tail

Because $\chi^2 \geq 0$, & only unusually large discrepancies b/w obs. & expected counts contradict H_0 .

Eg. $N(3.5, 0.7^2)$, $n = 40$

H_0 : battery follows $N(3.5, 0.7^2)$ distri

H_1 : battery does not follow

$$\chi^2 = 1.95 \quad k = 4, \quad v = 3$$

$$\chi^2 \sim \chi^2_3$$

$$\chi^2_{0.05, 3} = 7.815$$

$1.95 < 7.815$ fail to reject H_0

- χ^2 test for independence

H_0 : A & B are independent

H_1 : A & B are not "

independence, $P(A \cap B) = P(A) P(B)$

$$E_{ij} = \frac{(\text{row total})(\text{column total})}{\text{grand-total}}$$

$$\chi^2 = \sum_{\text{all cells}} \frac{(O_{ij} - E_{ij})^2}{E_{ij}}$$

$$df = (r-1)(c-1)$$

again; $\chi^2_{\text{calc}} > \chi^2_{0.05, v}$
reject H_0

- χ^2 test for homogeneity $\rightarrow$ suppose we have several populaⁿ in groups A, B, C. For each populaⁿ, indi. falls into categories like X & Y. We want to know if the categorical distribuⁿ look the same in all populations.

H_0 : category probab. are same across all populaⁿ

H_1 : at least 1 populaⁿ has diff.

$$\chi^2 = \sum \frac{(O - E)^2}{E}$$

$$df = (r-1)(c-1)$$

$$E_{ij} = \frac{(\text{row total})(\text{column total})}{n}$$

- 1 sample + 2 categorical var. $\Rightarrow$ independence
- several groups + 1 categorical responses $\Rightarrow$ homogeneity
- χ^2 statistic always non-negative
 $(O-E)^2$, but $(O-E)^2 \geq 0$
 $\& E > 0 \quad \therefore \frac{(O-E)^2}{E} \geq 0 \Rightarrow \chi^2 \geq 0$

we divide by E_i as this statistic measures discrepancy relative to the expected frequency.

- steps

1. 1 catego. variable vs known proportion
 $\Rightarrow$ goodness of fit
- 2 catego. variables on 1 sample $\Rightarrow$ indep.
 1 " " on multiple popula $\Rightarrow$ homogeneity
2. H_0/H_1
3. $E \rightarrow \chi^2 \rightarrow df \rightarrow p\text{-value/critical value}$
4. decide